The Elliptic-Curve Discrete Logarithm Does Not Know Which Curve It Is On: Pollard Rho Across Whole Isogeny Classes of Prime, Binary and Koblitz Curves

(with isogenies of degree 409 and 1721 found through torsion abscissae)

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Abstract

An $\mathbb{F}_q$-rational isogeny of degree prime to n restricts to an isomorphism of the prime-order subgroup $\langle P\rangle$ on which an elliptic-curve discrete-logarithm instance $Q = kP$ lives, so a generic algorithm should cost the same on every curve of the isogeny class. We test this in the only way an empirical statement about "which curve" can be tested: by carrying one instance, through explicit and verified isogenies, to many curves of its class and running the same Pollard rho on each, under a protocol frozen before the first measurement, with negative controls (the same curve in other coordinates or another field basis, an A/A replica), planted effects of known size, and the structural folds a crater curve admits as positive controls. Four classes are measured: a prime-field class of 347,004 curves (104 measured, spread over all 12 endomorphism-ring strata and reached through isogenies of degree 3 to 61), a $j = 0$ class (45 curves), a class of 327,080 curves over $\mathbb{F}_{2^{37}}$ (40 curves) and the class of the Koblitz curve $y^2+xy = x^3+1$ over $\mathbb{F}_{2^{41}}$ (708,153 curves, 11 measured over all four strata). In 85,425 solves, every one returning the planted logarithm, no curve, endomorphism-ring stratum, volcano level, isogeny degree, kernel-field degree, choice of generator (100 or 64 per curve with the same secret) or source curve of the instance left the preregistered Bonferroni band, and no across-curve test detected heterogeneity; the planted negation fold (0.63 to 0.77 of the unfolded mean, expected 0.707) and the 4-adding walk (1.27 to 1.36) were detected every time. The only real effects are the crater curves' own endomorphisms: the order-6 automorphism fold at $j = 0$ cost 0.439 $[0.401, 0.481]$ of the plain walk (expected $1/\sqrt{6} = 0.408$), and the Frobenius fold on the Koblitz crater 0.155 $[0.141, 0.170]$ (expected $1/\sqrt{41}$), 0.113 with negation (expected $1/\sqrt{82}$), still 1.7 to 2.2 times faster in wall time at 41 extra squarings per step. Every isogeny leads away from the one curve that has them. On the way we record that, for odd m, every prime dividing the conductor of a curve over $\mathbb{F}_{2^m}$ is $\pm 1 \bmod 8$, so the Koblitz curve over $\mathbb{F}_{2^{41}}$ has no isogeny off its crater of degree below 409; we find its 409- and 1721-isogenies (division polynomials of degree 83,640 and 1,480,920) in about one and five minutes each through ℓ-torsion abscissae in $\mathbb{F}_{2^{41\cdot 204}}$ and $\mathbb{F}_{2^{41\cdot 215}}$, and we derive and verify an explicit Vélu map on points for ordinary curves in characteristic two. This is a null result with passing positive controls, rho against rho; it is not an index-calculus result and makes no claim about any deployed curve. All code, records and raw per-solve rows are released.

1 Introduction

Isogenous elliptic curves over a finite field have the same number of points, and the isogeny class of an ordinary curve $E/\mathbb{F}_q$ is large: it is a union of endomorphism-ring strata, one for each order $\mathcal{O}$ between $\mathbb{Z}[\pi]$ and the maximal order $\mathcal{O}_K$ of $K = \mathbb{Q}(\pi)$, each a torsor under the class group of $\mathcal{O}$ [10, 4, 16]. It is natural to ask whether some curve in the class is a *better* place to solve a discrete logarithm than another, and the question has two very different answers depending on what "solve" means. For a *generic* algorithm the answer is a theorem (Section 2): the cost depends only on the group $\langle P\rangle$, which an isogeny of degree prime to n carries isomorphically. For algorithms that use a curve's structure the answer is yes in at least two known ways, both on the *crater*: the automorphisms of $j \in \{0, 1728\}$ fold rho by $\sqrt{3}$ or $\sqrt{2}$ [2, 21], and the Frobenius of a Koblitz curve folds it by $\sqrt{m}$ [7, 21]. Jao, Miller and Venkatesan [8] proved that the discrete logarithm is random self-reducible across an isogeny class under a rapid-mixing assumption on the isogeny graph, so an easier curve, if one existed, would make the whole class easy.

Related work, and what is and is not new. Galbraith [5] showed how to construct isogenies between ordinary curves of the same order and so transport a discrete logarithm between them; its cost for the worst pair of curves was improved recently [6], and Jao, Miller and Venkatesan [8] proved random self-reducibility under conditions on the endomorphism rings. These are reductions and algorithms for *moving* an instance; they do not measure what a generic solver costs on the far side, which is the question here. In a literature search we found no earlier measured whole-class comparison of rho with explicit isogenies; that is a statement about our search, not a proof that none exists. The null itself is a short consequence of the group law and is not offered as new. The Vélu map on points in characteristic two (Appendix A) is a specialisation of Vélu's characteristic-free formulas [19] to one model, and we make no claim that it was unknown; the mod-8 proposition of Section 4 is elementary and may well appear in the literature on isogenies of binary curves, where we did not find it stated.

None of this says what a measurement would show, and a measurement is what a claim of the form "curve X solves faster" or "no curve solves faster" needs. Such claims are comparative: one number on one curve has no reference. This paper is the measurement, done with the discipline a comparative claim requires: a frozen protocol, a null population, positive controls that prove the detector can see an effect of the size claimed absent, one instrument on every arm, and verdicts fixed by preregistered decision rules.

Contributions.

1. A native tool, `isoclass-rho`, that walks the ℓ-isogeny class of a curve over $\mathbb{F}_p$ or $\mathbb{F}_{2^m}$ over any set of primes, keyed by j-invariant, finds each ℓ-kernel by one of three routes (Elkies' construction from Φ_ℓ; factoring the division polynomial and running x-only ladders in the kernel's field $\mathbb{F}_{q^d}$; and, for degrees whose division polynomial is out of reach, a *torsion route* through ℓ-torsion abscissae in $\mathbb{F}_{q^d}$, $d = \mathrm{ord}(\pm\lambda)$), pushes points through the isogeny with Kohel's normalised formula over $\mathbb{F}_p$ and an explicit formula we derive for ordinary curves in characteristic two, verifies every edge on points, computes every stratum's class number and the exact class size, assigns each curve its endomorphism-ring conductor, and runs Pollard rho with negation, automorphism and Frobenius folds (Section 3).

2. A preregistered controlled experiment on four isogeny classes: prime, $j = 0$, generic binary and Koblitz, 200 measured curves and 85,425 solves, with isomorphic and field-basis controls, an A/A replica, two planted effects and three structural folds (Sections 5 and 6).

3. The result: within the stated resolution (about 18% per curve, 6.5% per stratum and 2% class-wide), nothing about the curve, its stratum, its level in any volcano, the route by which it was reached, the generator of $\langle P\rangle$ or the curve the instance was drawn on changes the cost of a generic walk, while the crater curves' folds are measured at their predicted sizes, down to 0.113 of the plain walk on the Koblitz crater (Section 6).

4. Two side results: for odd m every prime of the conductor of a curve over $\mathbb{F}_{2^m}$ is $\equiv \pm 1 \pmod 8$ (Proposition 1), which is why the Koblitz curve over $\mathbb{F}_{2^{41}}$ has no isogeny off its crater of degree below 409; and the torsion route that finds such 409- and 1721-isogenies in minutes (Section 4).

What this paper is not. It is not an attack, not an index-calculus result, and not a statement about any standardised curve. Its positive effects are the well-known crater folds, measured as positive controls; its main content is a null result whose detector demonstrably works.

2 The null is a theorem; what could still differ

Let $E/\mathbb{F}_q$ be ordinary, $P \in E(\mathbb{F}_q)$ of prime order n, $Q = kP$, and $\varphi\colon E \to E'$ an $\mathbb{F}_q$-rational isogeny of degree ℓ with $\ell \nmid n$. Then φ restricted to $\langle P\rangle$ is injective (its kernel has order ℓ), so $\varphi(P)$ has order n and $\varphi(Q) = k\,\varphi(P)$: the instance moves to E' with the same logarithm. A *generic* algorithm—one that touches group elements only through the group law and an injective encoding—has an expected cost that depends only on $(\langle P\rangle, P, Q)$ up to isomorphism, so its expected cost on E' equals its cost on E. For Pollard rho with an r-adding walk [14, 17, 18] the expectation is about $\sqrt{\pi n/2}$ group operations, times roughly $1/\sqrt{1-1/r}$ (1.016 at $r = 32$), and times $1/\sqrt{|G|}$ when the walk is folded by a group G of cheap maps on $\langle P\rangle$ whose orbits it canonicalises.

What can differ between curves is therefore only (i) the cost of a group operation, which depends on the coordinate representation and the field, not on the curve's position in the class; and (ii) the cheap maps a curve admits: $\mathrm{Aut}(E)$ beyond ±1 at $j \in \{0, 1728\}$, and the 2-power Frobenius $\tau\colon (x, y) \mapsto (x^2, y^2)$ when E is defined over $\mathbb{F}_2$. Both live on the crater: a curve with $\mathrm{Aut}(E) \neq \pm1$ has $\mathrm{End}(E) = \mathcal{O}_K$ for $K = \mathbb{Q}(\sqrt{-3})$ or $\mathbb{Q}(\sqrt{-4})$, and a curve whose coefficients lie in $\mathbb{F}_2$ has $\mathrm{End}(E) \ni \tau$ with $\mathbb{Z}[\tau] = \mathcal{O}_K$ for $K = \mathbb{Q}(\sqrt{-7})$. Every other curve of the class has lost the map (its j is not special; its coefficients are not in $\mathbb{F}_2$), while the generic cost is unchanged. The experiment below is designed to measure both halves of this statement.

3 The tool

`isoclass-rho` (`crypto:src/bin/isoclass_rho/`, about five thousand lines of Rust, no computation in Python) has two phases. `walk` explores the class and carries instances; `measure` runs rho on the recorded curves. Every statistic in this paper is computed in the tool; the plotting and tabulating scripts only lay out recorded values.

3.1 Class arithmetic

$\#E(\mathbb{F}_q)$ is counted by baby-step giant-step on random points and checked on eight more; $t = q+1-\#E$, $D = t^2 - 4q = f^2 D_K$. The class number $h(D_K)$ is counted from reduced primitive forms; $h(d^2 D_K)$ for every $d \mid f$ follows from the conductor formula and is cross-checked by counting $h(f^2 D_K)$ directly (the two agreed in every class). The class size is $\sum_{d|f} h(d^2 D_K)$, one curve per $\mathbb{F}_q$-isomorphism class in each stratum [10, 16].

3.2 Three routes to an ℓ-kernel

A prime ℓ is *active* when the curve has a rational ℓ-isogeny: $\ell \mid f$, or $\ell \nmid f$ with $\left(\frac{D_K}{\ell}\right) = 1$. All three routes produce the same object, the monic kernel polynomial h of degree $(\ell-1)/2$ whose roots are the abscissae of the non-trivial kernel points.

Elkies route ($\mathbb{F}_p$, $j \notin \{0, 1728\}$). The roots of $\Phi_\ell(j, Y)$ are the neighbours' invariants, and Elkies' construction [3, 15] gives h from a root; Φ_ℓ is built natively from the q-expansion of j ($\ell = 61$ takes 15 s). At $j \in \{0, 1728\}$ the automorphisms identify quotients of non-rational subgroups, so the roots of $\Phi_\ell(j, Y)$ over-count; there the next route is used.

Division route (any field, any j). The ℓ-division polynomial is factored over $\mathbb{F}_q$ into its irreducible factors of degree $d \mid (\ell-1)/2$; a root $x(T)$ of such a factor lies in $\mathbb{F}_{q^d}$, the x-only ladder gives $x(iT)$ there, and $\langle T \rangle$ is rational exactly when $\prod_i (x - x(iT))$ has coefficients in $\mathbb{F}_q$. The kernel-field degree d is the "hardness" of the kernel: the smallest extension its abscissae generate. In characteristic two we use Koblitz's recursion for $y^2 + xy = x^3 + ax^2 + b$ [9], checked against $x(3P) = x + f_2 f_4 / f_3^2$ on random points.

Torsion route (Section 4), for $\ell \mid f$ at a vertex above the floor, where the Frobenius is a scalar λ on $E[\ell]$ and every subgroup is rational.

3.3 Maps on points, and verification

Over $\mathbb{F}_p$ the normalised isogeny is Kohel's [10]

$$X = \ell x - 2\sigma - 2(3x^2 + a)\frac{h'}{h} - 4(x^3 + ax + b)\Big(\frac{h'}{h}\Big)', \qquad Y = y\,X'(x),$$

with σ the sum of the roots of h. Over $\mathbb{F}_{2^m}$, for $E\colon y^2 + xy = x^3 + ax^2 + b$ and odd ℓ, we specialise Vélu's general formulas [19, 20] to $a_1 = 1$, $a_3 = a_4 = 0$: with $L = h'/h$ and the Hasse derivatives $D_2 h$, $D_3 h$,

$$X = x + xL + x^2 L^2, \qquad Y = y + x^3 L_3 + xL + (x + y + x^2)(xL^2 + L), \qquad L_3 = L^3 + L\,\frac{D_2 h}{h} + \frac{D_3 h}{h}, \tag{1}$$

onto the codomain $y^2 + xy = x^3 + ax^2 + (b + \sigma + \sigma^2)$. The derivation (Appendix A) uses only that in characteristic two $\sum_Q 1/(x + x_Q)^2 = L^2$ and that the power sum $\sum_Q 1/(x + x_Q)^3$ follows from Newton's identities with the Hasse derivatives as elementary symmetric functions. Every edge the walk uses is then verified on points: images lie on the codomain, $\varphi(P + Q) = \varphi(P) + \varphi(Q)$, and $n\,\varphi(P) = O \neq \varphi(P)$;

$\#E$ is re-counted on every measured curve; and every rho solve must return the planted k. A wrong formula cannot survive these checks, which is how the characteristic-two map and the torsion route were validated in the tool's self-test before any measurement.

3.4 Levels, strata and instances

A curve's level in the ℓ-volcano [4] is found from its number of rational ℓ-kernels (one at the floor) and non-backtracking walks from three neighbours to the floor; its conductor relative to the walked primes is $\prod_\ell \ell^{\mathrm{level}_\ell}$, which names its endomorphism-ring stratum. The root's instance (P, kP) is pushed down the breadth-first tree to every measured curve. *Shifted* instances are drawn on other source curves and carried up to the root through dual edges (any rational isogeny back to the parent's j, composed with the model isomorphism, sixth roots at $j = 0$) and down again. *Generators* are $P_i = r_i P$, $Q_i = r_i Q$ with r_i uniform, the same k.

3.5 Rho and its folds

Single walker, 32-adding walk [17], distinguished points with $\theta = 2^{-8}$ [18], restart after $40 \cdot 2^8$ distinguished-point-free steps; trial s uses the same seed on every arm, and all (arm, seed) jobs are shuffled over the threads. The walk reproduces an earlier single-volcano tool's walk exactly. A *fold* steps from the canonical representative of a point's orbit under negation, $\mathrm{Aut}(E)$ at $j \in \{0, 1728\}$, or $\langle\tau\rangle$ (with or without negation) on a Koblitz curve, tracking the scalar; 2-cycles are prevented by the look-ahead rule and longer fruitless cycles are escaped deterministically from the cycle's minimum by doubling [21, 1].

4 The torsion route, and why the Koblitz class needs it

Proposition 1. *Let $E/\mathbb{F}_{2^m}$ be ordinary with m odd and conductor f, i.e. $t^2 - 2^{m+2} = f^2 D_K$. Every odd prime $\ell \mid f$ satisfies $\ell \equiv \pm 1 \pmod 8$.*

Proof. $\ell \mid f$ implies $t^2 \equiv 2^{m+2} \pmod{\ell^2}$, so $2^{m+2} = 4 \cdot 2^m$ is a square modulo ℓ, hence so is 2^m, hence (as m is odd) so is 2; and 2 is a quadratic residue modulo an odd prime ℓ exactly when $\ell \equiv \pm 1 \pmod 8$. □

So a binary curve over an odd-degree field has no 3-, 5-, 11- or 13-isogeny off any crater; the first candidates are $7, 17, 23, 31, 41, \ldots$ For the Koblitz curve $E_0 \colon y^2 + xy = x^3 + 1$ over $\mathbb{F}_{2^{241}}$, $D = -7 \cdot 703889^2$ with $703889 = 409 \cdot 1721$ and $h(-7) = 1$: the curve is alone on the crater of both volcanoes, every horizontal edge is a loop, and the only way off the crater is a 409- or a 1721-isogeny (`crypto:research/isogeny-class-rho/runs/run-20261018T0857Z/K/walk.json`, `class`). The division polynomials have degree 83,640 and 1,480,920, and even Φ_{409} is out of practical reach here.

The torsion route uses the structure instead. At a vertex above the floor of the ℓ-volcano the Frobenius acts on $E[\ell]$ as a scalar $\lambda \equiv t/2 \pmod \ell$, so every one of the $\ell + 1$ subgroups is rational and the abscissae of $E[\ell]$ lie in $\mathbb{F}_{q^d}$ with $d = \mathrm{ord}(\pm\lambda)$ in $(\mathbb{Z}/\ell)^* / \pm 1$ ($d = 204$ for $\ell = 409$, 215 for $\ell = 1721$). We build $\mathbb{F}_{q^d} = \mathbb{F}_q[y]/(g)$ with g an irreducible trinomial or pentanomial over $\mathbb{F}_2$ of degree d, coprime to m so it stays irreducible over $\mathbb{F}_{2^m}$ [11]. A random abscissa $x \in \mathbb{F}_{q^d}$ belongs to a point of E or of its quadratic twist over $\mathbb{F}_{q^d}$, whose orders $q^d + 1 \mp s_d$ follow from the Frobenius eigenvalue recurrence; the Montgomery x-only ladder [13, 12], which needs no y and does not distinguish a curve from its twist, by the matching cofactor gives an ℓ-torsion abscissa $x(T)$, checked by $x(\frac{\ell+1}{2}T) = x(\frac{\ell-1}{2}T)$. Its multiples $x(iT)$ follow from the differential-addition ladder, and the kernel polynomial is the product over Frobenius orbits $\{x(\pm i\lambda^j T)\}$, each orbit's polynomial having $\mathbb{F}_q$ coefficients (a non-rational subgroup, or a wrong λ, fails this test). One 409-kernel costs about one minute on a loaded core (a ladder of 8,347 bits in $\mathbb{F}_{2^{241 \cdot 204}}$ takes 31 s) and one 1721-kernel about five; the walk sampled three kernels of each degree at each vertex above the floor (54 ladders in all) and found the dual of a sampled edge by pushing a second torsion abscissa through it. Each of the ten resulting edges passed the point checks of Section 3, and $\#E$ was re-counted on all eleven curves.

5 Experimental design

The protocol (`crypto:research/isogeny-class-rho/PROTOCOL.md`, sha256 `81c3232b...`) was frozen before the first measurement and names the classes, the primes, the budgets, the controls, the decision rules, the multiple-comparison correction, the power check and the pre-flight runtime.

Classes. Table 1. P1 and B1 were found by random search for a conductor divisible by the wanted primes; P0 by a search among $j = 0$ curves; K is the Koblitz curve E_0 at the first field degree in $[23, 62]$ with a prime-order subgroup of at least 40 bits.

Table 1: The four classes. n is the prime order of $\langle P \rangle$; class size is $\sum_{d|f} h(d^2 D_K)$; "measured" counts the curves rho ran on, "strata" the endomorphism-ring strata they cover. Records: `crypto:research/isogeny-class-rho/runs/run-20261018T0857Z/<class>/walk.json`.

class	field; root curve	$D = f^2 D_K$	n (bits)	$h(D_K)$	class size	measured / strata
P1	$\mathbb{F}_p$, $p = 235212858169$; $a = 228505936335$, $b = 82673460332$	$315^2 \cdot (-8205107)$	35	612	347,004	104 / 12 of 12
P0	$\mathbb{F}_p$, $p = 217710133177$; $a = 0$, $b = 205801627284$ ($j = 0$)	$27792^2 \cdot (-3)$	33	1	38,472	45 / 12 of 20
B1	$\mathbb{F}_{2^{37}}$, $z^{37}+z^6+z^4+z+1$; $a = 89938361934$, $b = 61869205585$	$833^2 \cdot (-660703)$	37	296	327,080	40 / 6 of 6
K	$\mathbb{F}_{2^{41}}$, $z^{41}+z^3+1$; $y^2 + xy = x^3 + 1$	$703889^2 \cdot (-7)$	40	1	708,153	11 / 4 of 4

Primes walked. P1: $\ell \in \{2, \ldots, 19\}$ everywhere and $\{23, \ldots, 61\}$ at the first three vertices (the horizontal "hard" edges, with kernel fields up to $\mathbb{F}_{p^{23}}$); 3 (2 levels), 5, 7 (1 level each) are its volcano primes. P0: $\{2, 3, 5, 7, 11, 13\}$ (volcanoes of depth 4 and 2; the prime 193 of its conductor was not walked). B1: $\{7, 17\}$ everywhere (7 of depth 2, 17 of depth 1), $\{23, 31\}$ at the first three vertices. K: $\{409, 1721\}$ by the torsion route. Inert primes with $\ell \nmid f$ were recorded as inactive.

Measured set. The root, up to eight curves per stratum spread over the breadth-first order, and the curves reached through the largest degrees. The root's instance was carried to each.

Modes and budgets. *Class*: every measured curve × 100 paired seeds, plus three isomorphic models of the root ($\mathbb{F}_p$: $(u^4 a, u^6 b)$; $\mathbb{F}_{2^m}$: the same curve in three other polynomial bases of the field, since $y \mapsto y + sx$ leaves every abscissa and hence the walk unchanged), an A/A replica of the root under an independent seed set, a planted negation fold (expected 0.707), a planted 4-adding walk (Teske's heuristic $\sqrt{1 - 1/32}/\sqrt{1 - 1/4} = 1.137$), and the structural folds where the root admits them. *Generators*: 6/4/6/3 curves × 100/100/64/64 generators × 25 seeds. *Shift*: 5/4/5/3 sources × every measured curve × 25 seeds.

Decision rules. R1: a curve against the pooled solves of the others by Welch's z, family m = number of curves, $\alpha = 0.05$, threshold z_{crit} at α/m two-sided; R2: Kruskal–Wallis and one-way ANOVA across curves with the between-curve variance ratio var(arm means)/(pooled var/T), heterogeneity "detected" if $p < 0.05$; R3: the same for pooled strata, levels, distances, parent degrees and kernel-field degrees; R4: the planted negation fold's interval must exclude 1 and the structural folds must lie within 15% of $1/\sqrt{|G|}$, else every null verdict of that class is inconclusive; R5: per curve, Kruskal–Wallis across generators, the variance ratio, Bonferroni-flagged generators and Spearman correlations of the generator mean with r, bits(r), popcount(r) and $x(P_i)$; R6: Kruskal–Wallis by target and by source and η^2 of each factor. Verdict wording is fixed: "within the control distribution" or "more extreme than the family". Power: with a single-solve coefficient of variation of 0.52, $T = 100$ gives a per-curve mean with 5.2% standard error, so a per-curve effect of about 18% is detectable at $m \approx 100$, a stratum effect of about 6.5%, a class-wide shift of about 2%; the strong plant (29%) should be seen per arm at $|z| \approx 5.6$, the weak plant (14%) near the threshold.

Execution. The walks were run from a frozen command card by a separate executor; the measurements by the author's session; every command, log and record is in the run directory. Each class mode was replayed at five seeds afterwards and reproduced the recorded iteration counts exactly (550/550, 260/260, 230/230, 95/95 rows). Wall times come from a shared four-vCPU virtual machine with other jobs running and are indicative only; group operations are the unit of every comparison.

6 Results

6.1 Every curve, stratum and route

Table 2: Class mode: 100 paired seeds per arm, all solves correct. "ratio" is the pooled mean over $\sqrt{\pi n/2}$ ($\times 1.016$ expected for the 32-adding walk); $|z|_{\max}$ is the largest Welch z of a measured curve against the pooled others, with the Bonferroni threshold; VR is the between-curve variance ratio. Records: `crypto:research/isogeny-class-rho/runs/run-20261018T0857Z/<class>/class/results.json`.

class	curves	solves	$\sqrt{\pi n/2}$	pooled mean	ratio	$\lvert z\rvert_{\max}$ / z_{crit}	KW p	ANOVA p	VR
P1	104	10,400	202,614	206,675	1.020	3.03 / 3.49	0.089	0.101	1.18
P0	45	4,500	110,515	110,942	1.004	2.42 / 3.26	0.162	0.264	1.13
B1	40	4,000	328,549	334,995	1.020	3.02 / 3.23	0.158	0.292	1.11
K	11	1,100	929,277	924,231	0.995	2.38 / 2.84	0.209	0.076	1.70

In every class every measured curve is within the control distribution (Table 2), and no across-curve test detects heterogeneity. The pooled distributions are the familiar rho shape: for P1, minimum 2,082, median 193,529, maximum 705,972 operations, coefficient of variation 0.52, skewness 0.64. The spread of curve means is 1.1 to 1.7 times what the pooled single-solve spread predicts (11,751 against 10,814 on P1; 62,858 against 48,378 on K's eleven curves), the largest ratio on the smallest family and carried by no single curve. Per-step cost is flat within a class (429–523 ns per operation across P1's 104 curves, 528–600 ns across B1's 40).

Table 3: P1 pooled by endomorphism-ring stratum: conductor f_v of $\mathrm{End}(E_v)$ relative to $\mathcal{O}_K$, the stratum's class number $h(f_v^2 D_K)$ (how many curves share that ring), eight measured curves each, 800 solves. Record: `crypto:research/isogeny-class-rho/runs/run-20261018T0857Z/P1/class/results.json`, `groups`.

f_v	$h(f_v^2 D_K)$	mean $\pm$ 95% CI	ratio	z vs rest
1 (crater)	612	212,144 $\pm$ 7,338	1.047	1.52
3	1,224	206,365 $\pm$ 5,391	1.019	−0.12
5	3,672	206,804 $\pm$ 7,578	1.021	0.03
7	4,896	208,763 $\pm$ 7,647	1.030	0.56
9	3,672	212,672 $\pm$ 7,651	1.050	1.60
15	7,344	208,215 $\pm$ 7,594	1.028	0.41
21	9,792	201,608 $\pm$ 7,458	0.995	−1.39
35	29,376	201,269 $\pm$ 7,395	0.993	−1.49
45	22,032	200,839 $\pm$ 7,461	0.991	−1.60
63	29,376	204,788 $\pm$ 7,453	1.011	−0.52
105	58,752	206,945 $\pm$ 7,220	1.021	0.08
315 (floor)	176,256	209,999 $\pm$ 7,352	1.036	0.92

Table 3 shows the twelve strata of P1, from the crater (End $= \mathcal{O}_K$, 612 curves) to the floor (End $= \mathbb{Z}[\pi]$, 176,256 curves): all within $\pm 5\%$ of the reference with $|z| \leq 1.6$. The same holds for every level of every volcano in every class, for the distance from the root, for the degree of the edge that reached the curve (3 to 61 on P1, 7 to 31 on B1, 409 and 1721 on K) and for the kernel-field degree (1 to 23 on P1, 3 to 11 on B1, 204 and 215 on K); the curves reached through P1's hard horizontal isogenies have ratios 0.88–1.09 with $|z| \leq 3.0$, B1's 23- and 31-isogeny targets 0.92–1.09, and K's four strata 0.996 (the Koblitz crater), 1.009, 1.001 and 0.978 ($|z| \leq 0.8$). The curves on the far side of a 1721-isogeny cost what the Koblitz curve costs.

6.2 Controls, planted effects and the crater folds

Table 4: Control and fold arms on each root (100 seeds): mean ratio to $\sqrt{\pi n/2}$ with the bootstrap 95% interval, Welch z against the class's pooled isogenous solves, and mean wall time per solve. Isomorphic controls are three (u^4a, u^6b) models (P1, P0) or three other field bases (B1, K). Records: `crypto:research/isogeny-class-rho/runs/run-20261018T0857Z/<class>/class/results.json`.

class	arm	expected	ratio [95% CI]	z	ms / solve
P1	iso controls	1	1.158 [1.059, 1.268]; 1.076; 1.052	2.58; 1.00; 0.54	104–114
P1	A/A replica	1	1.053 [0.967, 1.148]	0.70	103
P1	plant: negation fold	0.707	0.725 [0.651, 0.804]	−7.45	77
P1	plant: 4-adding walk	1.137	1.298 [1.172, 1.440]	4.03	122
P0	iso controls	1	1.067; 1.054; 1.023	1.17; 0.87; 0.40	53–58
P0	A/A replica	1	0.964 [0.878, 1.059]	−0.83	52
P0	plant: negation fold	0.707	0.634 [0.567, 0.709]	−9.98	36
P0	plant: 4-adding walk	1.137	1.265 [1.136, 1.411]	3.67	62
P0	**fold:** Aut **of** $j = 0$	$1/\sqrt{6} = 0.408$	**0.439 [0.401, 0.481]**	−25.3	26
B1	basis controls	1	1.043; 0.897; 1.039	0.47; −2.55; 0.35	174–199
B1	A/A replica	1	1.161 [1.049, 1.285]	2.23	217
B1	plant: negation fold	0.707	0.771 [0.702, 0.847]	−6.40	147
B1	plant: 4-adding walk	1.137	1.300 [1.177, 1.435]	4.20	256
K	basis controls	1	0.983; 1.029; 0.963	−0.20; 0.61; −0.55	457–625
K	A/A replica	1	0.970 [0.875, 1.078]	−0.45	456
K	plant: negation fold	0.707	0.769 [0.705, 0.838]	−5.92	375
K	plant: 4-adding walk	1.137	1.358 [1.221, 1.508]	4.74	650
K	**fold: Frobenius τ**	$1/\sqrt{41} = 0.156$	**0.155 [0.141, 0.170]**	−48.2	275
K	**fold: τ and negation**	$1/\sqrt{82} = 0.110$	**0.113 [0.103, 0.124]**	−53.1	212

Table 4 is the calibration. The isomorphic and field-basis controls and the A/A replicas sit inside the band ($|z| \le 2.6$): a change of coordinates or of seeds moves a 100-seed mean exactly as much as any isogenous curve does, which is the point of the null population. The planted negation fold is detected in every class at $|z|$ between 6 and 10, and the planted 4-adding walk at $|z| \approx 4$; the latter comes out larger (1.27–1.36) than Teske's $1/\sqrt{1 - 1/r}$ heuristic predicts for $r = 4$, as a walk that far from random should. The structural folds come out where theory puts them: the order-6 automorphism fold on the $j = 0$ crater at 0.439 against 0.408 (within the 15% tolerance of R4), the Frobenius fold on the Koblitz crater at 0.155 against 0.156 and, with negation, 0.113 against 0.110. The Koblitz folds cost 1.92 and 2.05 µs per step against 0.50 (forty-one squarings to find the orbit's minimum) and still win in wall time, 275 and 212 ms per solve against 459 ms. Neither fold exists on any other curve of its class: Aut $= \pm 1$ away from $j = 0$, and τ is an endomorphism only when the curve is defined over $\mathbb{F}_2$, which only the crater curve is. The plain walk costs the same on all of them.

6.3 Generators and source curves

On P1, 100 generators of $\langle P \rangle$ on each of six curves with the same secret, 25 seeds each (15,000 solves): the standard deviation of the 100 generator means is 19,852–21,863 against 21,275–21,873 expected from the single-solve spread (variance ratios 0.84–1.04), Kruskal–Wallis p between 0.44 and 0.92 per curve, no generator beyond the threshold (family 600, largest $|z|$ 3.76 against 3.93), and every Spearman correlation of the generator mean with r, bits(r), popcount(r) or $x(P_i)$ has $|\rho| \le 0.12$. P0 (400 generators), B1 (384) and K (192) read the same (variance ratios 0.81–1.26 per curve), with one cell of 384 on B1 beyond the band on the low side (0.677 [0.535, 0.829] at 25 seeds, Mann–Whitney $z = -3.3$), the low tail of 384 small means and not repeated by that curve's other 63 generators or by its features. Which generator carries the instance does not matter.

Instances drawn on five source curves of P1 (the root and four others) and carried to all 104 targets, 25 seeds per cell (13,000 solves): the two-way decomposition gives $\eta^2 = 0.0085$ for the target and 0.0002 for the source, Kruskal–Wallis $p = 0.19$ by target and 0.37 by source, and the five sources' pooled means are 1.007–1.029 of the reference. Two of the 520 cells lie beyond the Bonferroni band on the low side; both targets are unremarkable in the other four rows and in the class mode (1.013, 1.036), so these are the low tail of many 25-solve means. P0, B1 and K read the same ($\eta^2_{\text{source}} \le 0.0012$ everywhere). Where the instance was drawn does not matter either.

6.4 End to end: the only route-dependent cost is finding the isogeny

Carrying an instance to a curve costs the isogenies on the way; solving it costs the same everywhere. On P1 the Elkies edges cost 0.6 ms ($\ell = 3$) to 5.5 s ($\ell = 61$, dominated by building Φ_{61} once) and a solve 98 ms; the transport to any measured curve was under 74 ms. On B1 the 23- and 31-isogenies cost 1.3–2.4 s each by the division route against a 0.19 s solve. On K the 409- and 1721-isogenies cost 197 s and 978 s per three-kernel search, so the transport to a conductor-703889 curve is nine to fifteen minutes against a 0.46 s solve. The 3D heat maps of the run (Figures 1 and 2) colour each measured curve by its mean operations and by its mean end-to-end time; the first is flat, the second is a map of the transport.

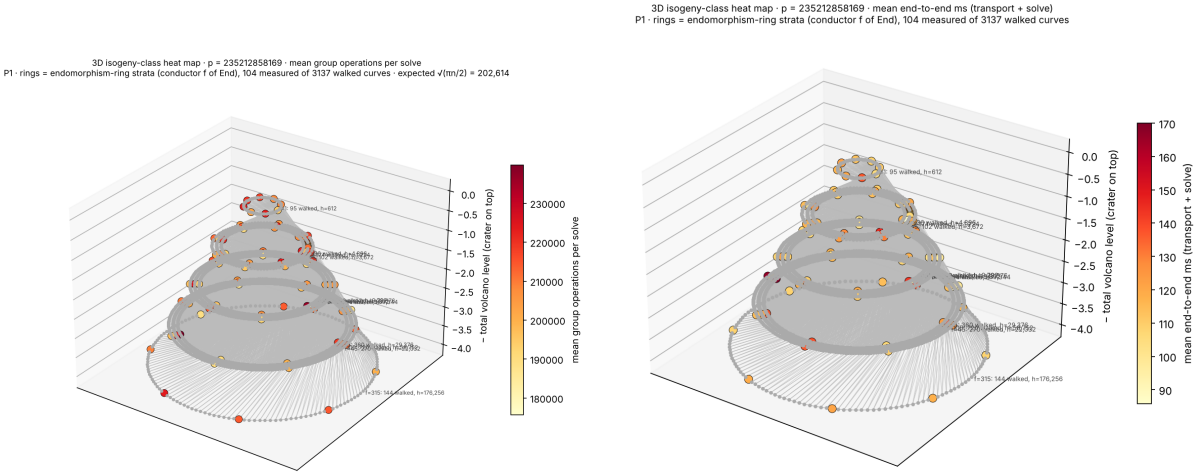


Figure 1: P1: the 3,137 walked curves on their endomorphism-ring strata (rings, crater on top, with each stratum's class number), the 104 measured ones coloured by mean group operations (left) and by mean end-to-end time, transport plus solve (right). Record: `crypto:research/isogeny-class-rho/runs/run-20261018T0857Z/P1/`.

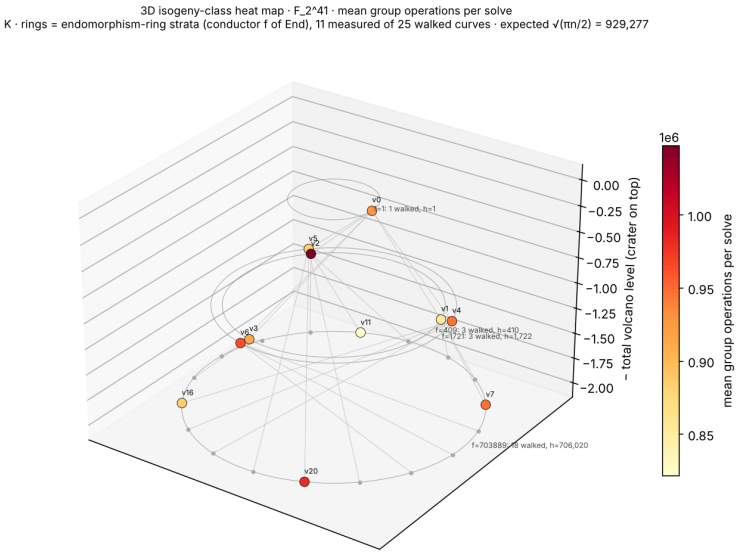


Figure 2: K: the Koblitz crater, its 409- and 1721-strata and the conductor-703889 stratum of 706,020 curves; 11 measured of 25 walked, coloured by mean operations. Record: `crypto:research/isogeny-class-rho/runs/run-20261018T0857Z/K/`.

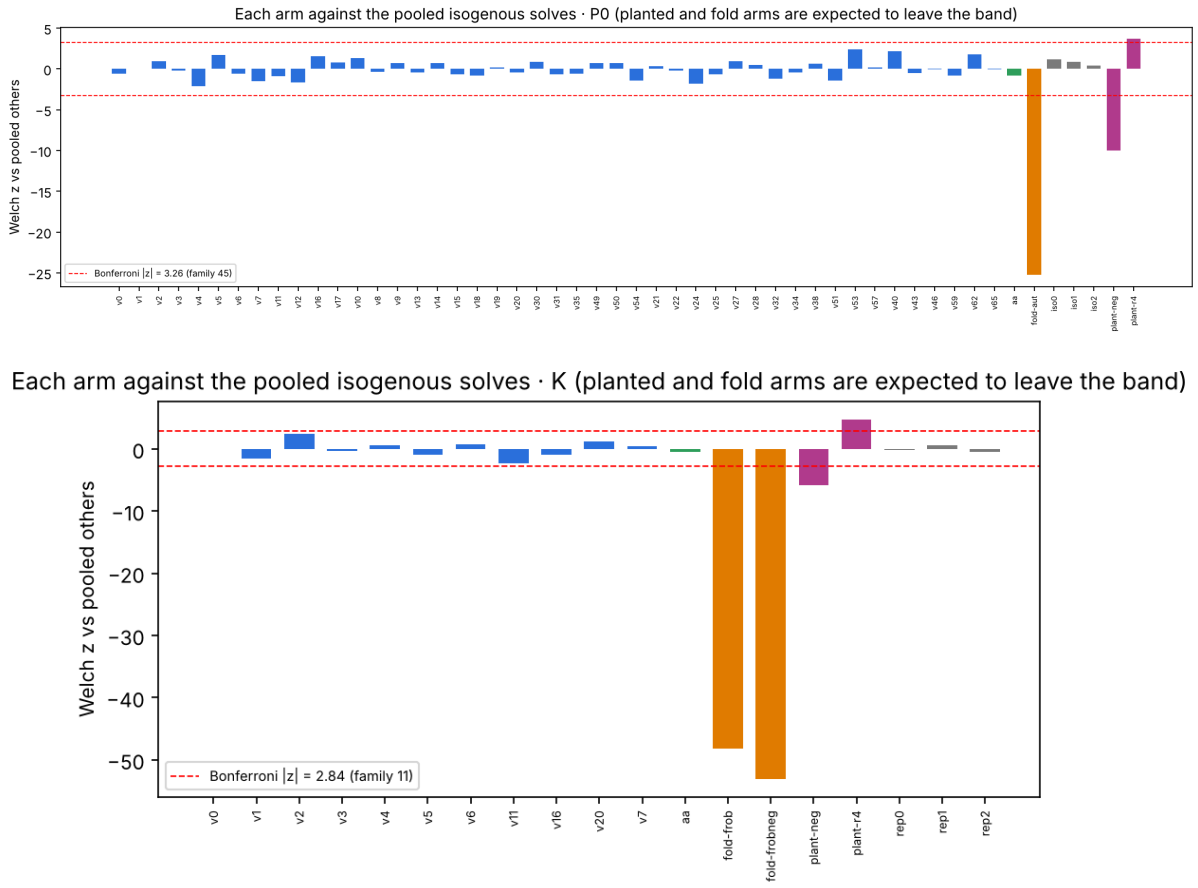


Figure 3: Each arm against the pooled isogenous solves, P0 (top) and K (bottom): every curve and control inside the Bonferroni band, the planted and structural folds outside it. Records: `crypto:research/isogeny-class-rho/runs/run-20261018T0857Z/P0/`, `crypto:research/isogeny-class-rho/runs/run-20261018T0857Z/K/`.

7 Discussion

What the result says. Within the resolution the budgets buy, the position of a curve in its isogeny class is invisible to a generic walk: not the endomorphism-ring stratum, not the level in any volcano, not the degree or kernel field of the isogeny that reached it, not the generator, not the curve the instance came from. The effects that do exist are the two the theory names, the crater's automorphisms and the Koblitz Frobenius, and they are measured at their predicted sizes. The experiment therefore supports the operational reading of random self-reducibility [8] from the other side: not only can an instance be moved anywhere in the class, nothing is gained generically by moving it, and the one thing that can be lost is the crater's structure.

What it does not say. It does not say that no curve of a class is easier for some non-generic method; index calculus on a curve with a convenient model, or a transfer to another group, is not a generic walk and is not measured here. It does not bound effects below the stated resolution. It does not speak about curves outside these four classes or at cryptographic sizes, where the only extrapolation we would make is the theorem of Section 2. And the folds are not "isogeny speed-ups": they are the crater curve's own endomorphisms, which an isogeny takes away.

Hard isogenies. The Koblitz class is the instructive case. Its reachable neighbours are at degrees 409 and 1721 because Proposition 1 forbids small ones, and division polynomials of degree 10^5–10^6 put the naive route out of reach; the torsion route reaches them in minutes because above the floor the Frobenius is a scalar and the kernel field is as small as $\mathrm{ord}(\pm\lambda)$ allows. The curves on the far side are generic curves

over $\mathbb{F}_{2^{41}}$ with the Koblitz curve's group and none of its structure: in practice, a 1721-isogeny is a way to turn a Koblitz instance into a non-Koblitz one, and nothing else.

Classification. In the vocabulary of the companion repository this is a null result with passing positive controls, rho against rho on one instrument. It is not a change to any index-calculus baseline and is not a speed-up claim.

8 Reproducibility

Everything is in the companion repository: the tool (`crypto:src/bin/isoclass_rho/`, with a self-test that checks every formula of Section 3 against direct computation, including the characteristic-two map, the torsion route against the division route and the duals), the frozen protocol, the command card, every walk record (models, levels, conductors, kernel polynomials, carried instances, transport costs), every per-solve row (arm, seed, iterations, restarts, distinguished points, escapes, wall time), the per-arm summaries and tests, the replays, the figures and the report. The run is `crypto:research/isogeny-class-rho/runs/run-20261018T0857Z`; the study README is `crypto:research/isogeny-class-rho/README.md`. The library's test suite (4,380 tests) passes at the commit of the run.

A The characteristic-two map on points

Vélu's formulas for a general Weierstrass model and a finite subgroup C [19, 20], with S a set of representatives of $(C \setminus \{O\})/\pm$ and no 2-torsion in C, read $X = x + \sum_{Q\in S}\left[v_Q/(x-x_Q) + u_Q/(x-x_Q)^2\right]$ and $Y = y - \sum_{Q\in S}\left[u_Q(2y+a_1x+a_3)/(x-x_Q)^3 + v_Q(a_1(x-x_Q)+y-y_Q)/(x-x_Q)^2 + (a_1u_Q - g_Q^x g_Q^y)/(x-x_Q)^2\right]$ with $g_Q^x = 3x_Q^2 + 2a_2x_Q + a_4 - a_1y_Q$, $g_Q^y = -2y_Q - a_1x_Q - a_3$, $v_Q = 2g_Q^x - a_1g_Q^y$, $u_Q = (g_Q^y)^2$, onto $A_4 = a_4 - 5v$, $A_6 = a_6 - (a_1^2 + 4a_2)v - 7w$, $v = \sum v_Q$, $w = \sum(u_Q + x_Qv_Q)$. For $y^2 + xy = x^3 + ax^2 + b$ in characteristic two: $g_Q^x = x_Q^2 + y_Q$, $g_Q^y = x_Q$, $v_Q = x_Q$, $u_Q = x_Q^2$, $w = 0$, so $A_4 = v = \sigma$ and $A_6 = b + \sigma$; the substitution $Y \mapsto Y + \sigma$ removes the A_4 term and gives the codomain $y^2 + xy = x^3 + ax^2 + (b + \sigma + \sigma^2)$. In X, with $u = x + x_Q$, $x_Q/u = 1 + x/u$ and $x_Q^2/u^2 = (x_Q/u)^2$, so $X = x + d' + xL + d'^2 + x^2L^2 = x + xL + x^2L^2$ since $d' + d'^2$ is even, where $L = \sum 1/u = h'/h$ and $\sum 1/u^2 = L^2$ in characteristic two. In Y the representative y_Q cancels (replacing y_Q by $y_Q + x_Q$ adds $2x_Q^2/u^2 = 0$), the terms reduce to $y + \sum\left[x^3/u^3 + x/u\right] + (x + y + x^2)\sum\left[x/u^2 + 1/u\right] + \sum x_Q$, and the final shift $Y \mapsto Y + \sigma$ cancels the $\sum x_Q = \sigma$. The remaining power sum $L_3 = \sum 1/u^3$ follows from Newton's identities, $p_3 = e_1p_2 + e_2p_1 + e_3$ in characteristic two, with $e_k = D_kh/h$ the Hasse derivatives (the coefficients of $h(x+T)/h(x)$), giving $L_3 = L^3 + L\,D_2h/h + D_3h/h$ and formula (1). The formula is checked in the tool's self-test by the homomorphism, codomain and order tests on random points for $\ell \in \{3, 5, 7, 11, 13\}$ on Koblitz and generic binary curves, and in the run by the same tests on all 533 binary edges.

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